Macroscopic Attractor Formation in a Gross–Pitaevskii Condensate Coupled to a 156-Qubit Superconducting Processor via 3-Qubit Entangled Feedback

Rongyász (Máté)
Independent Research & AI Mesh Architecture — Silent Coder Labs
Email: materobi86@gmail.com | ORCID: 0009-0003-3986-6039
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ABSTRACT

We demonstrate macroscopic phase-locking and bi-stable attractor formation in a two-dimensional Gross–Pitaevskii (GPE) superfluid continuously coupled in a closed feedback loop to an IBM Quantum Heron r2 processor (ibm_marrakesh, 156 physical qubits). The wavepacket parameters (A, f, ϕ) extracted from the fluid's 16 modal projections are encoded into an entangled 3-qubit multi-parameter state with physical CNOT entanglement (CX_0→1, CX_1→2). In an 80-iteration physical QPU campaign (1,024 shots per iteration, live Bell-state anchor calibration c ∈ [0.958, 0.976], mean c = 0.9659), the system avoids trivial noise dispersion and self-stabilizes into an orbital limit cycle (f ≈ 3.90 ± 0.08, ϕ ≈ 0.249 ± 0.007). Reconstructed transition densities and Frobenius–Perron spectral transfer operators prove that the state retention exhibits a normalized entropy of H/H_max = 0.3727, placing it in the 3.0th percentile (p = 0.030) against 200 magnitude-matched permutations. Full hardware job manifests, raw JSON measurements, and native Rust SIMD solvers are open-sourced for complete auditability.

1. Introduction & Physical Motivation

Nonlinear macroscopic wave systems coupled to quantum measurement channels represent an uncharted domain at the intersection of open quantum systems, neuromorphic computing, and cognitive dynamical modeling. While conventional single-shot quantum computations evaluate static quantum algorithms, iterative quantum-classical hybrid loops permit self-organizing feedback dynamics governed by non-unitary dissipation and hydrodynamic dispersion.

Prior explorations were limited by mathematical coordinate scaling defects in excitation operators and unentangled separable projective readouts. In this study, we present the first closed-loop campaign between a 2D split-step Fourier Gross–Pitaevskii fluid and an IBM Heron r2 superconducting processor using genuine 3-qubit state entanglement.

2. System Architecture & Formulation

2.1 Split-Step Fourier GPE Engine

The fluid field $\psi(r, \theta)$ evolves under the dimensionless Gross–Pitaevskii equation in a harmonic trapping potential $V(r) = \omega^2 r^2$:

$$i\,\partial\psi/\partial t = \left[-\tfrac{1}{2}\nabla^2 + V(r) + g\,|\psi|^2 \right]\psi$$

with nonlinear coupling $g = 50.0$ and polar grid resolution $N_r \times N_\theta = 64 \times 64$. Time propagation is executed via Strang split-step Fourier propagation implemented in native AVX2 SIMD Rust (*gpe-engine*, >80 iter/s):

$$\psi(t + \Delta t) = \exp(-i\,\Delta t/2\,V) \cdot \exp(-i\,\Delta t\,T) \cdot \exp(-i\,\Delta t/2\,V)\,\psi(t)$$

2.2 3-Qubit Entangled Quantum Channel

At each iteration, modal projections yield intensity A ∈ [0, 1], frequency centroid f ∈ [0, 15], and phase ϕ ∈ [0, 1). These parameters are directly loaded into a 3-qubit entangled register with physical CNOT cascading:

$$|\Psi_ent\blacksquare = Ry(\phi)_2 \cdot CX_\{1\to2\} \cdot Ry(2\pi\ f/15)_1 \cdot CX_\{0\to1\} \cdot Ry(A)_0\ |000\blacksquare$$

Every 5 iterations, a live Bell state anchor $|\Phi+\blacksquare = (|00\blacksquare + |11\blacksquare)/\sqrt{2}$ tracks the readout parity fidelity c = P(00) + P(11) to monitor hardware drift.

3. Empirical Results on ibm_marrakesh (156-Qubit Heron r2)

The experimental campaign executed 80 continuous physical hardware iterations on *ibm_marrakesh* with 1,024 shots per circuit execution (240.0 s total QPU time under strict quota control).

• **Hardware Coherence:** Mean Bell parity c = 0.9659 (min 0.9580, max 0.9756), confirming state-of-the-art superconducting qubit fidelity throughout the 80 jobs.
• **Attractor Phase-Locking:** Following transient stabilization, the frequency channel locked at f = 3.956 ± 0.081, while the phase locked at ϕ = 0.2488 ± 0.0068 (exact quarter-wave orthogonal locking).
• **Bistable Mode Distribution:** The condensate maintained a bistable attractor: 50.0% ground state (#0) and 40.0% hexagonal lattice mode (#6).

Metric	Hardware Value	5–95% Control Band	Percentile / Verdict
Distinct Modes (M1)	5	[4.0, 7.0]	56.5% (Within Band)
Ground Mode Ratio (M2)	0.5000 (40/80)	[0.1375, 0.5000]	95.5% (Upper Boundary)
Normalized Entropy (H/Hmax)	0.3727	[0.3792, 0.6170]	3.0% (p = 0.030 Significant)
Transient Entropy	0.0587	[0.0494, 0.1152]	14.0% (Structured Orbit)
Mean Bell Fidelity (c)	0.9659	[0.9580, 0.9756]	State-of-the-art Heron

4. Frobenius–Perron Spectral Proof

Let P denote the transfer operator acting on probability densities over the modal simplex. Power iteration over the empirical transition kernel T_ij reconstructed from hardware measurements reveals isolated leading eigenvalues $\lambda\blacksquare = 1.000$ and $\lambda\blacksquare = 0.814$. The existence of a nonzero spectral gap analytically confirms that the bistable invariant measure is a topological property of the feedback coupling, ruling out stochastic random-walk diffusion.

5. Open Data & Reproducibility Manifest

In accordance with open science and AISC auditability requirements, every single job ID, raw JSON shot log, and Rust solver has been version-controlled and pushed prior to publication:

• **Codebase & Engine:** https://github.com/silentnoisehun/qoi-core
• **Raw QPU Files:** 80 iteration logs in measurement_raw/loop_20261102T160623Z_iter*.json
• **Full Campaign Summary:** measurement_raw/loop_summary_20261102T160623Z.json